

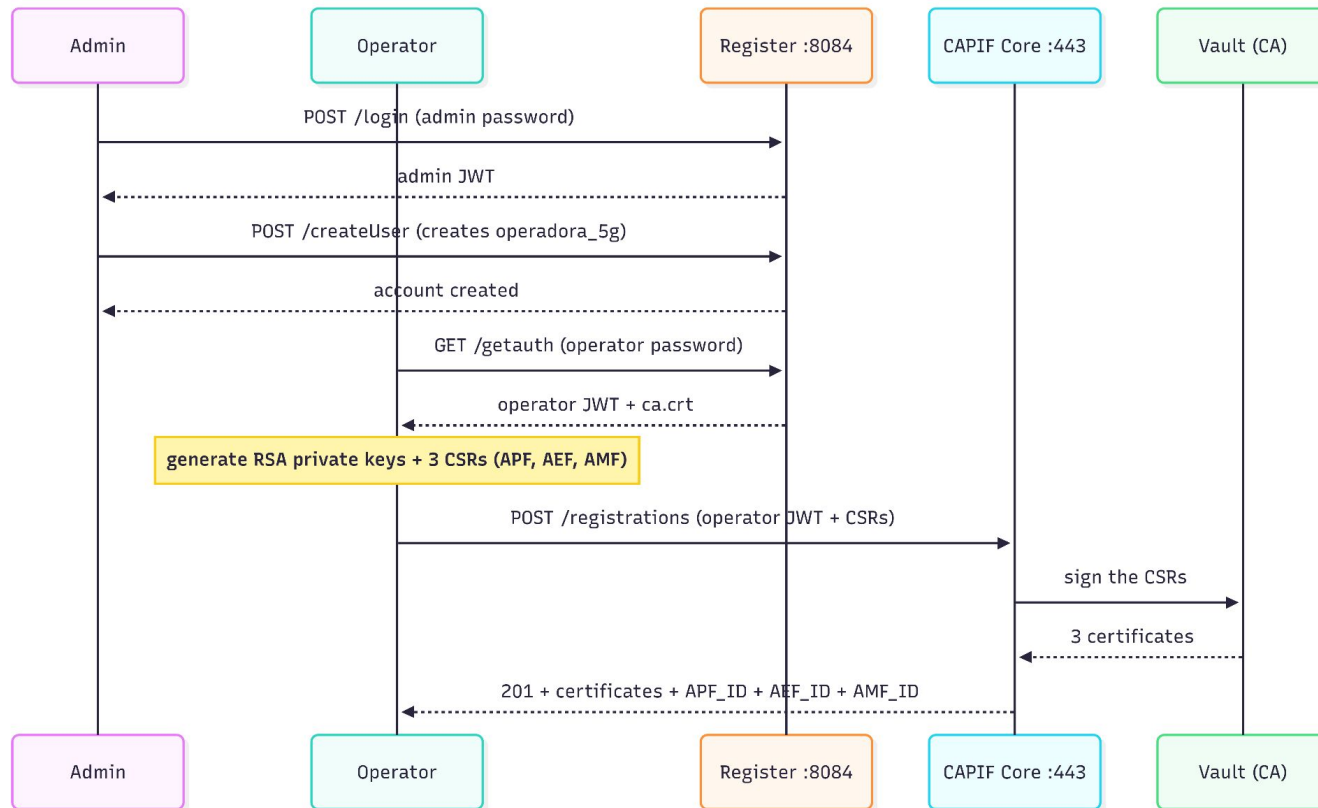
Exposing a SIM Swap API with OpenCAPIF



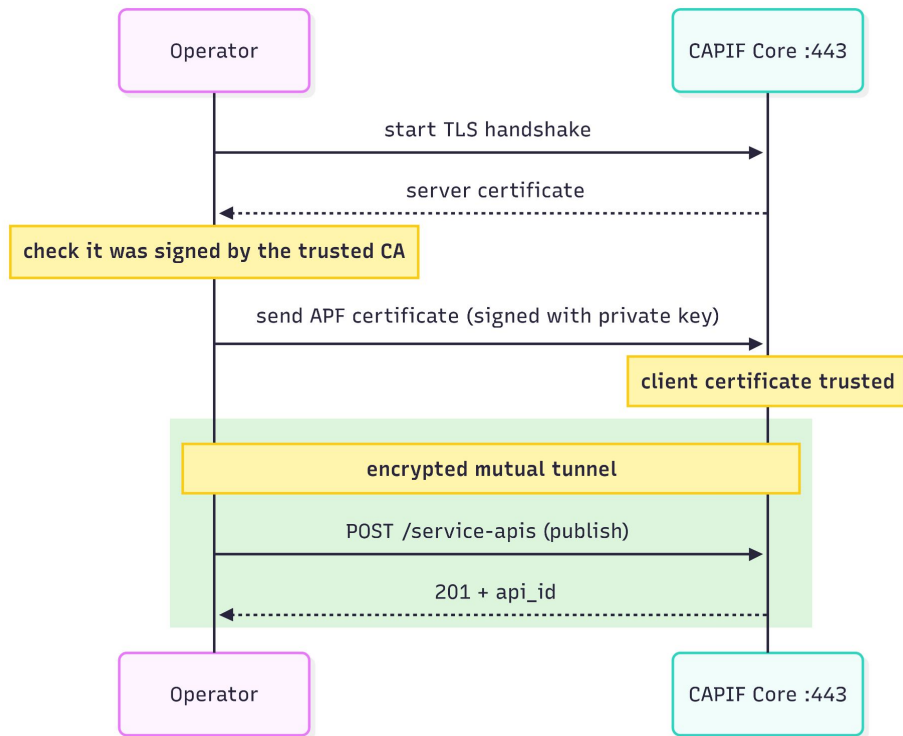
David Pelicano

24.06.2026

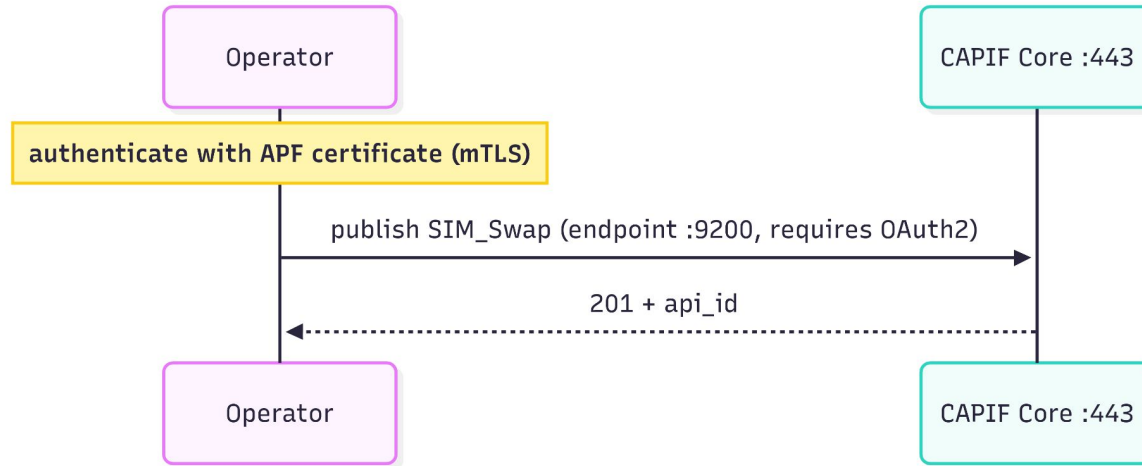
Operator Onboarding – obtaining the certificates



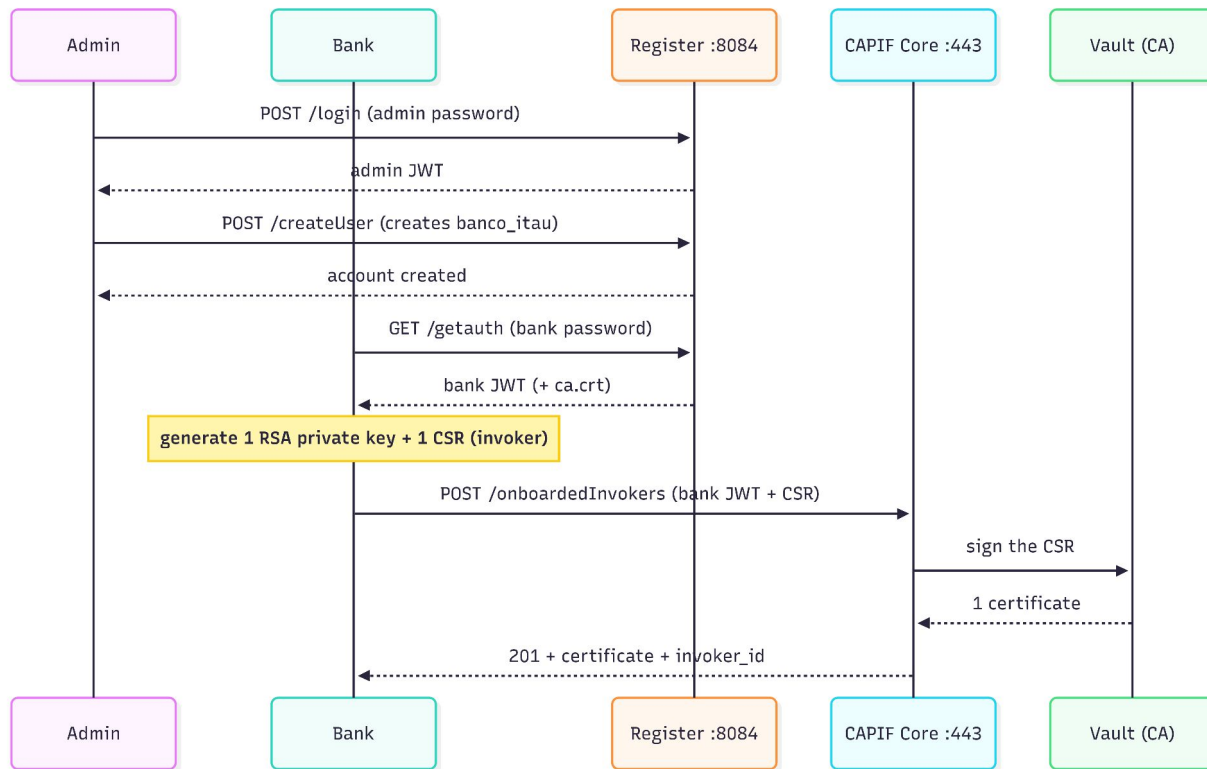
mTLS – Authentication with Certificates



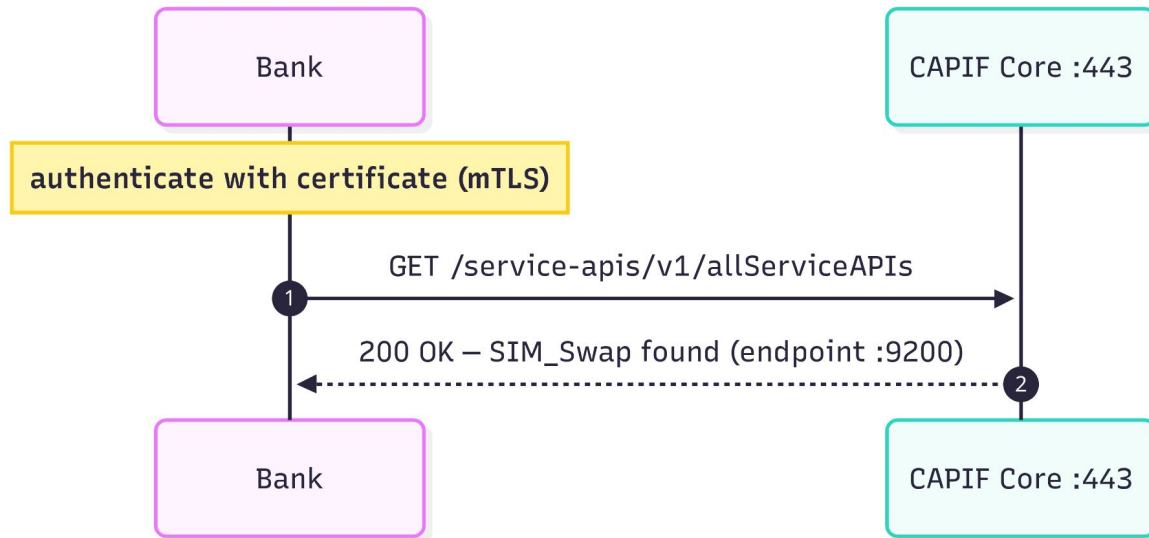
Operator Publishes the Network API (mTLS)



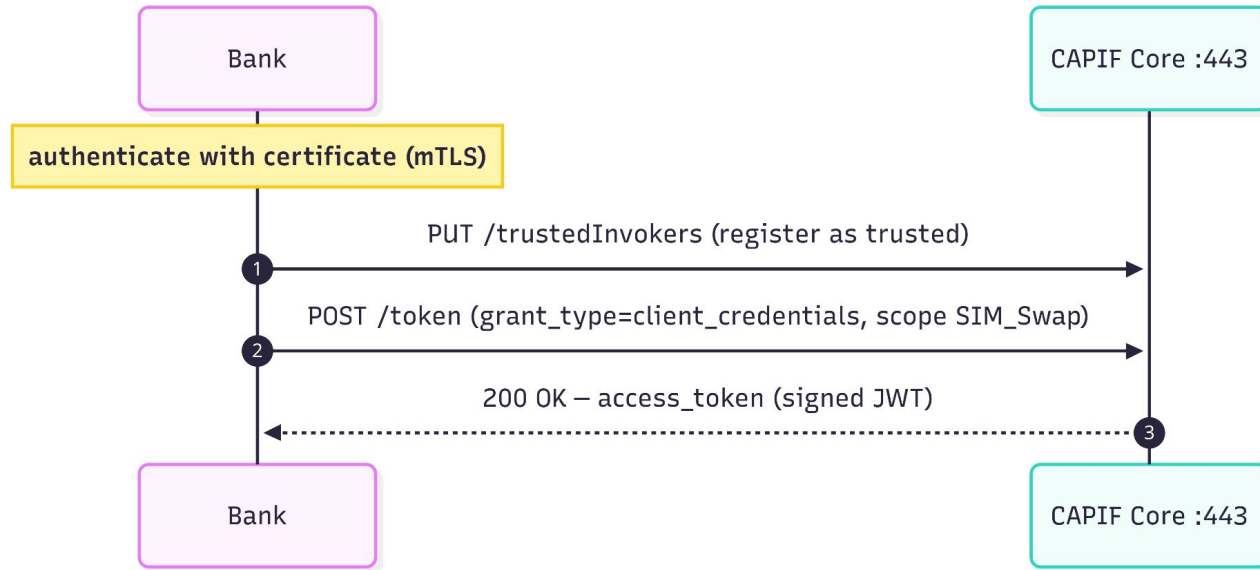
Bank Onboarding (Invoker)



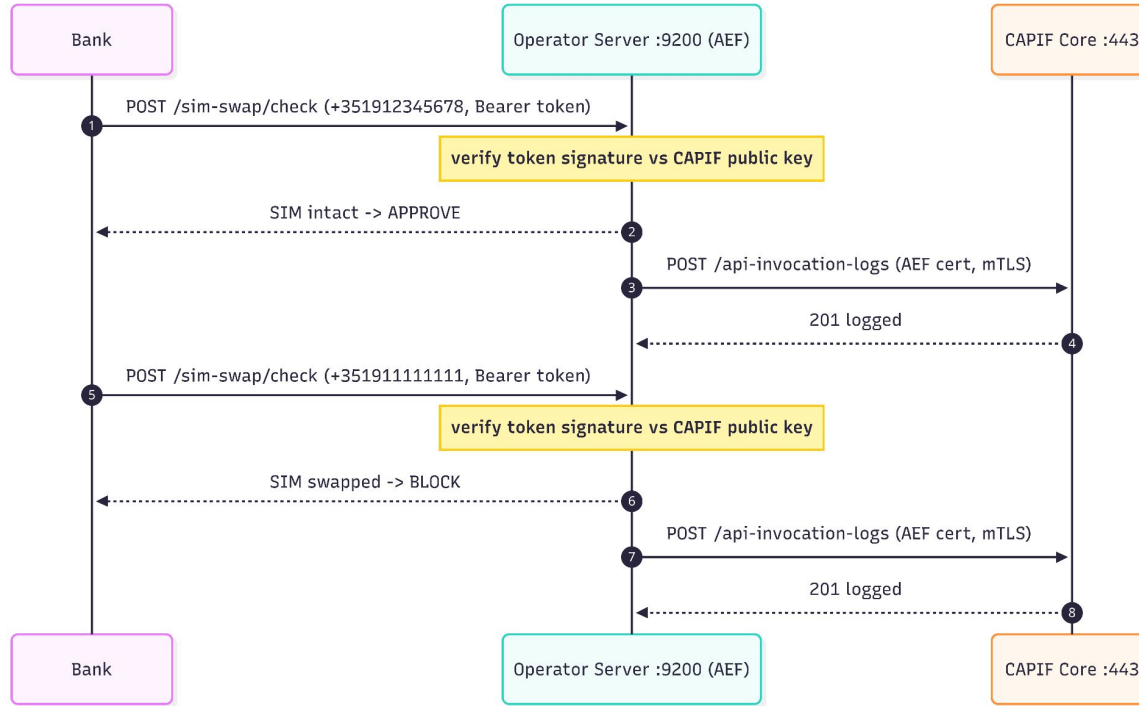
API Discovery



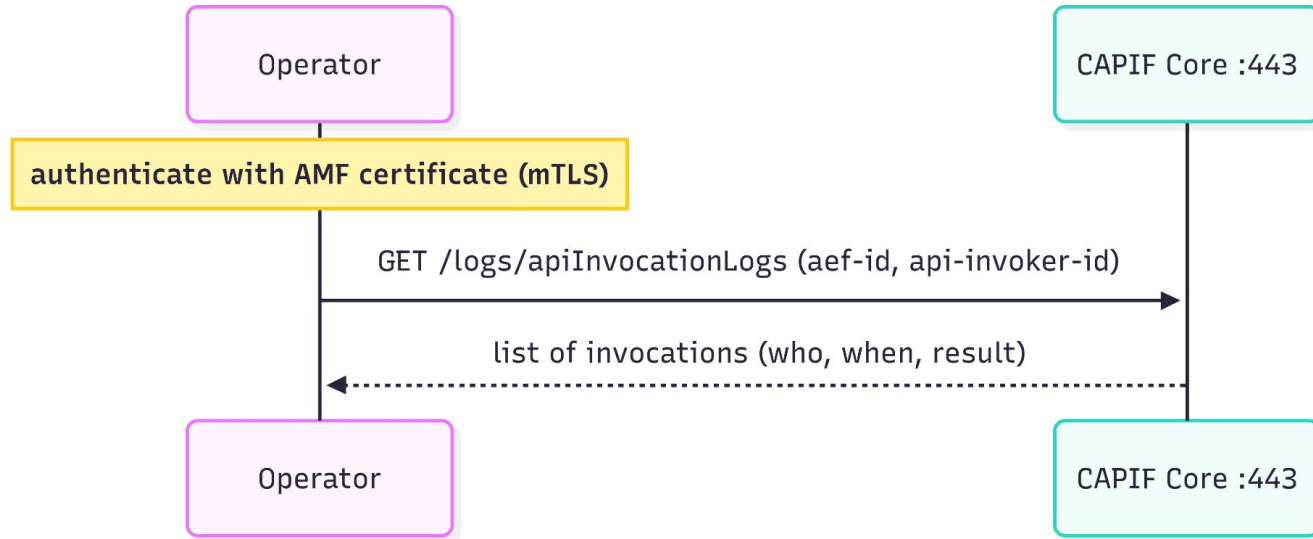
Getting the OAuth2 Token



The API Call: Fraud Check



Audit (AMF)



Real-World Use Case: SIM Swap API in Banking

The Problem: Brazil loses ~\$54B annually to digital scams involving SIM swapping and account takeover.

The Solution: Itaú Unibanco uses CAMARA APIs to verify SIM status, processing 36.5M security checks.

Impact: Enhanced customer protection; banks prevent fraud before transactions are approved.

Retrieve SIM swap date

POST /retrieve-date Retrieve SIM swap date

Check SIM swap

POST /check Check SIM swap



Itaú Unibanco combats payment scams

Case study for bank payments using the CAMARA-standardised SIM Swap API - [View API Description](#)

Business Problem

In Brazil, most banking transactions are now conducted through digital channels. These transactions are vulnerable to fraud, where attackers use stolen personal data and social engineering to hijack a user's SIM card and take control of their account. About US\$34 billion is lost to scams in Brazil each year according to the Global Anti-Scam Alliance.

Impact

Itaú Unibanco can now detect SIM card swaps early and quickly, identifying potential fraud and providing an extra layer of protection for its users. Between its adoption by Itaú Unibanco and April 2023, the SIM Swap API has been processed more than 36.5 million times, with a current monthly average of 5 million processes.

Technical Solution

To help automate payments, Itaú Unibanco, working with the leading mobile operators in Brazil, is employing the SIM Swap API to source a date and time stamp showing when a SIM associated with a phone number was last changed. The API is one of several CAMARA-based network APIs launched in Brazil.

Value

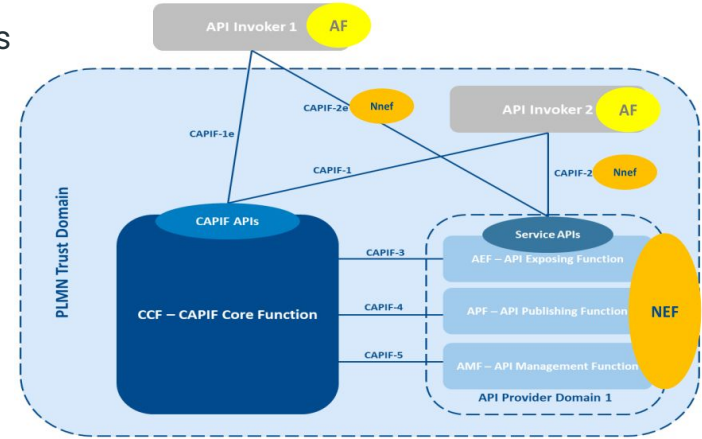
Itaú Unibanco's customers are benefitting from a more secure experience, and are less vulnerable to fraud, when using digital channels to make a transaction.

CAPIF – Common API Framework

- Problem: Each API had its own rules
- Solution: One framework with same rules for all APIs

- Roles

- API Invoker: consumes APIs
- API Provider: exposes APIs
- CCF: central controller



- Onboarding – registers and authorizes who can use/provide APIs (CCF)
- Discovery – finds available APIs (Invoker asks CCF)
- Auth – CCF gives token → Invoker sends token → Provider validates
- Logging – CCF records all API calls

Real-World Use Case: Envelope (CAPIF Implementation)

ENVELOPE Project: A 5G-Advanced platform designed for Connected and Automated Mobility (CAM)

- **Challenge:** CAM apps in untrusted domains need secure 5G API access

OpenCAPIF Implementation (3GPP TS 23.222)

- **Distributed API Providers:** Network APIs integrated across multiple domains, from Far-Edge (roadside) and Edge up to the Central Cloud.
- **CAPIF Core Function (CCF):** The “brain” that manages authentication and Topology Hiding, preventing external entities from seeing the internal network structure.
- **Secure API Invokers:** Mobility applications (CAM) running in untrusted domains that use CAPIF to discover and consume 5G services securely.

